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1 a

① a

$$t^* = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right)$$

$$= 6.62\times10^{-7} \exp\left(\frac{11949}{625}\right)$$

$$= 133.01 \text{ days}$$

$t = 500$ days $> t^*$

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$$\cancel{\delta_0 K_L t^{1/n}} \quad \delta^* = 5.1 \exp\left(-\frac{550}{T}\right)$$

$$= 2.11 \ \mu m$$

$$K_L = 7.48\times10^6 \exp\left(-\frac{12500}{T}\right)$$

$$= 0.015 \ \mu m/d$$

$$\delta = \delta^* + K_L (t - t^*)$$

$$= 2.11 + 0.015 \times (500 - 133.01)$$

$$= 7.61 \ \mu m$$

1b/

$$C_H^{clad}(\text{wt. ppm}) = \frac{2 f \times \delta \times \rho_{oxide} \times f^{O}_{ZrO_2} \times \frac{M_H}{M_O}}{\left(t - \frac{\delta}{PBR}\right) \times \rho_{metal}} \times 10^6$$

$$= \frac{2\times 0.14 \times 7.61 \times 5.68 \times \frac{32}{32+91} \times \frac{1}{16}}{\left(600 - \frac{7.61}{1.56}\right) \times 6.5} \times 10^6$$

$$= 30.87 \text{ wt. ppm}$$

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(2)

The rate limiting step in aqueous ~~or~~ corrosion of Zr clad is the diffusion of oxygen through the ZrO_2 layer.

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(3)

In usual operating condition hydrides form circumferentially around the cladding near outer ~~rad~~ radius forming a rim.

H_2 tends to travel down the temperature gradiant and accumulate to tensile stress area. So in operation condition the circumferentially hydrides form. In dry post irradiation conditions radial hydrides can form ~~as the~~ that are much more brittle.

Hydrides embrittle the Zr cladding leading to brittle fracture of the cladding and failure. In accident conditions it plays an important role in fuel failure by brittle fracture.

(4)

RIA is reactivity insertion accident where a high amount of reactivity is inserted into the core over a very short time period causing a rapid immediate power excursion locally.

In PWR a typical RIA is a ~~rod~~ control rod ejection (or withdrawal) accident were in BWR it is control rod drop accident.

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During a RIA as the control rod is ejected or dropped from the core, leads to a rapid power rise into the fuel which is suppresed by feedback effects. ~~and other~~ However the rapid power increase cause high thermal gradiants and high temperatures locally that causes the fision bubbles to grow and can cause fuel fragmentation. It can also increase the rod internal pressure. ~~Due to these effects~~, ~~The high fuel may file~~ temperature can also drastically oxidize the Zr cladding and sometime partially melt it. So in RIA fuel may fail through PCMI, oxidation, ballooning/burst or partial melting.

(5) LOCA is a loss of coolant accident were one of the main primary coolant line breaks and disrupt the core cooling. In a LOCA a break would cause the reactor to scram so the power generation halts, but the decay heat keeps heating up the fuel. As the cooling is lost and ECCS pumps water to cool down the reactor, the balooning or deformation of the fuel rods may hamper the coolant's capability to flow and cool the rods. ~~Besides due to~~ The balooning occurs because the coolant pipe break depressurizes the core while the rod internal pressure expands the fuel cladding outwards

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In LOCA this ballooning or burst can lead to fuel failure.
Other mechanisms include the steam oxidation of Zr cladding causing runaway reaction at high temperature where the steam is introduced by ECCS trying to cool the high temperature rods.

The main difference between LOCA and RIA is in the progression speed. ~~RIA~~ is LOCA is a much slower accident than RIA.

⑥ Burnup impacts the type of failure in several ways. The fuel pellet ~~expand~~ expansion due to burnup ~~can~~ leads to PCMI during an accident causing failure. The fission gas release over the burnup period ~~can~~ cause fuel internal pressure can lead to ballooning burst during a LOCA. The oxygen that is released due to fission travels to the cladding and can oxidize the Zr cladding internal surface. ~~The~~ The volatile fission gas such as Cs, I can cause ~~the~~ loss of strength in Zr cladding by ~~~~ chemically interacting and also by pressure rise. High burnup also means higher oxidation of the Zr outer surface and embrittlement that can also increase the probability of failure in accident conditions.

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7. The four pathways to make fuel system accident tolerant

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1. Improved reaction kinetics with steam
2. Improved cladding properties
3. Improved fuel properties
4. ~~Better~~ Enhanced fission product retention.

8. A near term ATF concept is to apply a ~~prot~~ corrosion resistant coating on the outer ~~clad~~ Zr cladding surface by transition metal or ceramic layer or MAX material (M = transition metal, A = Al, Si, X = C, N).

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This is of interest because it can prevent runaway steam oxidation of Zr at high temperature by forming a protective oxide layer on outer cladding surface through oxidation.

9. The ~~o~~oxidation of Zr is an exothermic reaction.

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At high temperature if the oxidation heat is not adequately ~~removed~~ removed by coolant or other measures, it can cause more oxidation to occure causing a runaway oxidation of Zr in which

-mostly

the whole Zr cladding layer can be oxidized and lost leading the failure.

(10)

Three limiting phenomenon in LWR

1. DNBR caused by CHF
2. Cladding oxidation and ~~embrittlement~~. Hydrogen pickup.
3. ~~fuel melting~~ Power to fuel melting.

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(11)

CRUD has very low thermal conductivity that leads to higher centerline temperature. It also can trap boron causing to CRUD induced power shifts in PWR. It may chip or flake off with activation products that may leads to radiation expose to the other primary coolant system components

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(12)

6/6

LiOH injection: Boric acid is used to control reactivity in PWR but it lowers the pH and can increase Zr oxidation. So LiOH is added to control the pH.

Zn injection: It makes a more stable spinel structure and prevent [illegible] out of the CRUD activation products leading to lower radiation exposure to components.

(13)

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In MOX fuels there forms a central void due to burnup. It is caused by the vapor transport up the temperature gradient where fuel vaporizes in high temperature and recrystallizes in low temperature region. It leads to a central void and a columnar grain region immediately outward.

Another phenomenon is JOG (Joint Oxide Gaine), where the fission product oxides accumulate in the fuel cladding gap and forms a buffer layer between the fuel and cladding.

This JOG layer has higher conductivity than gas gap ~~[illegible]~~ and can lead to lower fuel temperatures.